

Food Delivery Analytics Project Report

Exploratory Data Analysis, SQL Analytics, and Power BI Dashboard

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1 Executive Summary

This project demonstrates an end-to-end analytics workflow for a food-delivery operation. The source material contains a relational table named `food_delivery_orders` with order identifiers, dates and times, customer areas, restaurants, order values, preparation time, distance, delivery time, delivery partner, status, cancellation reason, weather, and traffic level. The analysis moves from raw records to decision-oriented outputs.

The exploratory analysis establishes the main operational patterns. Delivery time varies by area, and the supplied area chart shows that the ten displayed areas are relatively close to one another, with a spread of only a few minutes between the highest and lowest bars. In contrast, traffic has a much stronger visible relationship with delivery time: the average is approximately 82 minutes under severe traffic, compared with approximately 60 minutes under low traffic. Weather is also associated with cancellations. The supplied chart places heavy rain at roughly 15%, above rain, cloudy, and clear conditions.

The cancellation-reason chart identifies long delivery time as the largest reason, followed by delivery-partner unavailability and restaurant delay. This suggests that cancellation reduction requires a combined response across dispatch capacity, restaurant operations, and customer communication rather than a single intervention. The hourly demand chart shows that orders fluctuate across the full day, with a visible peak around midday and a trough around late morning.

The Power BI workbook contains a single report page with eleven visual containers: cards, bar charts, column charts, and a line chart. These visuals cover hourly demand, cancellation reasons, area-wise delivery time, weather and order status, restaurant delivery time, area-wise order value, traffic-level order volume, and traffic-level delivery time. Together, the workbook functions as a compact management dashboard.

2 Project Background and Business Context

Food-delivery platforms operate through a chain of time-sensitive activities. A customer places an order, a restaurant prepares it, a delivery partner collects it, and the order travels across a distance affected by traffic and weather. A failure at any stage can increase delivery time or produce a cancellation. For this reason, a single metric such as total orders is not enough to evaluate performance.

The present project uses analytics to connect customer demand with operational execution. It asks practical questions: Which hours create the greatest demand? Which reasons dominate cancellations? Which locations experience longer deliveries? How does traffic affect delivery time? Does weather change cancellation risk? Which restaurants may require process improvement? The SQL queries in the supplied project documentation are designed around these questions.

2.1 Problem Statement

The analytical problem is to transform order-level data into reliable, interpretable indicators that can support operational decisions. The project must preserve data quality, apply consistent filters, calculate meaningful averages and rates, and communicate the results through charts and a dashboard.

2.2 Objectives

The primary objective is to build a repeatable pipeline from data cleaning to dashboard reporting. The supporting objectives are to measure order volume, delivered and cancelled orders, cancellation rate, average delivery time, average order value, delivered revenue, cancellation reasons, area performance, hourly demand, traffic impact, weather impact, and restaurant-level performance.

2.3 Scope of the Analysis

The project focuses on descriptive and diagnostic analytics. It does not attempt to forecast future demand, optimize routes algorithmically, or estimate causal effects. All interpretations should therefore be read as patterns observed in the supplied data rather than proof that one variable alone causes another.

3 Dataset Description and Data Model

The central table is `food_delivery_orders`. Each row represents an order record, while the columns describe its identity, timing, commercial value, operational duration, status, and context. The design is suitable for both exploratory analysis in Python and aggregation in SQL or Power BI.

Field	Type	Analytical meaning
Order_ID	Text	Unique order reference used for counting records.
Order_Date	Date	Calendar date of the order.
Order_Time	Time	Time of day used for hourly demand analysis.
Customer_Area	Text	Customer location used for geographic comparison.
Restaurant	Text	Restaurant associated with the order.
Order_Value_INR	Decimal	Monetary value of the order in Indian rupees.
Food_Prep_Time_Min	Integer	Preparation duration in minutes.
Distance_KM	Decimal	Travel distance associated with the order.
Delivery_Time_Min	Integer	Delivery duration in minutes.
Delivery_Partner	Text	Assigned delivery-partner category or identifier.
Order_Status	Text	Delivered, cancelled, or another recorded status.
Cancellation_Reason	Text	Reason captured when an order is cancelled.
Weather	Text	Weather condition at the time of the order.
Traffic_Level	Text	Traffic classification for the order.

The SQL schema uses appropriate types for the project. Dates and times are stored separately, numerical durations are stored as integers, monetary values and distance use decimal precision, and categorical attributes use text fields. This structure makes it possible to use functions such as `HOUR()`, `AVG()`, `SUM()`, `COUNT()`, `CASE`, `GROUP BY`, and `ORDER BY` in a transparent manner.

3.1 Data Grain and Metric Discipline

The grain of the dataset is the order. Counts should therefore count order records or non-null order identifiers, while averages should be calculated only on records for which the relevant measure is meaningful. The supplied SQL correctly filters delivered orders when calculating average delivery time and delivered revenue. Cancellation analysis, on the other hand, uses all orders so that the denominator represents the complete order population.

4 End-to-End Methodology

The project follows a staged workflow. First, the raw data is inspected and cleaned in the exploratory analysis environment. Second, the cleaned dataset is loaded into a relational database and queried using SQL. Third, the analytical results are connected to Power BI, where they are presented as interactive or semi-interactive visuals.

Raw Data → Cleaning → EDA → SQL → Power BI → Insights

4.1 Stage 1: Data Preparation and EDA

The EDA stage is used to understand column names, data types, missing values, duplicate records, category consistency, and the range of numerical values. It also creates derived fields such as `Order_Hour` from `Order_Time`. The project charts are generated with Python plotting code and saved as image files for later use.

4.2 Stage 2: SQL Analysis

The SQL stage creates a database and table, validates the schema, checks the number of records, and calculates business metrics. SQL is especially useful for repeatable aggregation: the same query can be rerun after new data is loaded, and the output can be connected to a reporting layer.

4.3 Stage 3: Power BI Dashboard

Power BI is used to combine summary cards and comparative visuals in one page. The PBIX layout indicates eleven visuals using a red-and-black premium theme. The dashboard therefore acts as the presentation layer, while Python and SQL act as the analytical preparation layers.

4.4 Quality Principles Applied

The report uses delivered-only filters for delivery-time averages, all-order denominators for cancellation rates, explicit sorting for rankings, and clear labels for minutes, percentages, order counts, and rupee values. These choices reduce the risk of mixing incompatible populations.

5 Data Cleaning and Preparation

Data cleaning is not merely a technical precondition; it determines whether later metrics are trustworthy. A delivery-time average that includes cancelled orders, a cancellation rate with a missing denominator, or a duplicated order identifier can materially distort the dashboard.

5.1 Recommended Cleaning Checks

The project's cleaning stage should verify that `Order_ID` is unique or that duplicate records are understood, that date and time fields parse correctly, that numerical columns are within reasonable ranges, and that category labels use consistent spelling and capitalization. Empty cancellation reasons should be reviewed for delivered orders, while cancelled orders should be checked for missing reasons.

Check	Why it matters	Treatment in analysis
Missing values	Missing categories or measures can bias rates and averages.	Inspect, document, and handle according to business meaning.
Duplicates	Repeated orders can inflate counts and revenue.	Check order identifiers and remove confirmed duplicates.
Invalid durations	Negative or implausible minutes indicate data errors.	Review ranges before aggregation.
Category consistency	Different spellings split the same category.	Standardize labels before grouping.
Time parsing	Incorrect hour extraction changes demand patterns.	Convert time values before using <code>HOUR()</code> .
Status logic	Delivered and cancelled populations require different filters.	Use explicit <code>Order_Status</code> conditions.

5.2 Derived Field: Order Hour

The hourly analysis groups records by the hour extracted from `Order_Time`. The resulting `Order_Hour` field ranges from 0 through 23 and supports demand monitoring across the day. This field is reused in both the Python line chart and the SQL query that orders hourly demand.

5.3 Cleaning as a Reproducible Step

A strong project submission should preserve the cleaning notebook or script alongside the cleaned data. Reproducibility means that another analyst can understand what was changed, why it was changed, and how the final dataset was produced. The cleaning stage should therefore contain comments, before-and-after row counts, and a short data-quality summary.

6 Exploratory Analysis I: Area Delivery Time



图 1 Average delivery time by customer area.

The area chart compares the average delivery time for delivered orders across ten customer areas. The bars are visually close, indicating that area-level variation exists but is not as large as the variation associated with traffic. OMR appears at the upper end of the displayed range, while Ambattur appears at the lower end. The result suggests that location should be monitored, but area alone may not explain the entire delivery experience.

One possible operational interpretation is that the platform’s service footprint is broadly consistent across these areas. Small differences may arise from distance, restaurant density, road connectivity, delivery-partner allocation, or time-of-day mix. The correct next step is not to label an area as poor based on the average alone, but to drill into distance, traffic, restaurant, and hour for the same area.

6.1 Analytical Implication

Area comparisons are most useful when accompanied by sample size and a consistent delivered-order filter. A dashboard can improve this view by adding a tooltip with order count, average distance, average preparation time, and cancellation rate. This would distinguish a genuinely slow area from an area with a small number of unusually long deliveries.

6.2 SQL Logic Used

The supplied SQL groups delivered orders by Customer_Area, calculates the average Delivery_Time_Min, and orders the result from highest to lowest average. This makes the query directly suitable for identifying priority areas.

7 Exploratory Analysis II: Cancellation Reasons

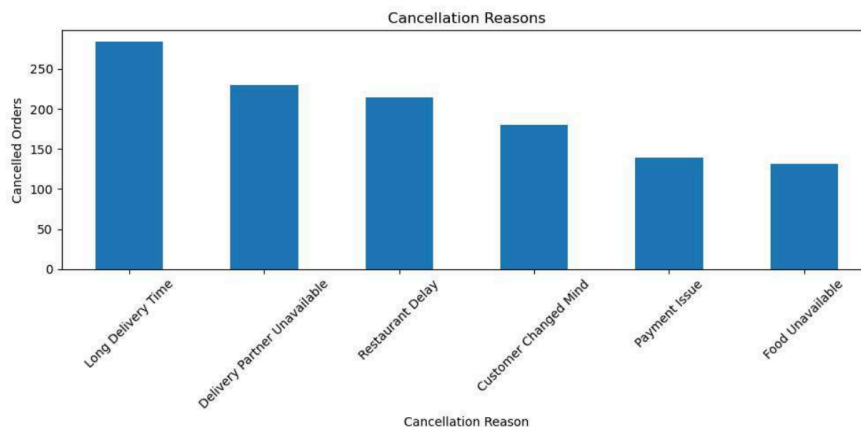


图 2 Distribution of cancellation reasons.

The cancellation-reason chart shows that **Long Delivery Time** is the dominant recorded reason. **Delivery Partner Unavailable** is the second-largest category, followed by **Restaurant Delay**. The remaining reasons are customer changed mind, payment issue, and food unavailable. The ranking is important because it turns a general cancellation problem into a set of operational categories.

The first three categories are primarily connected to execution. Long delivery time may be influenced by traffic, distance, preparation delay, or dispatch capacity. Delivery-partner unavailability points to supply-demand imbalance in the delivery fleet. Restaurant delay suggests that kitchen preparation or order acceptance is a bottleneck. Together, these categories indicate that cancellation reduction should begin with time reliability.

7.1 Recommended Diagnostic Split

The cancellation chart should be cross-filtered by hour, traffic, weather, restaurant, and area. For example, if long-delivery cancellations rise sharply during severe traffic, the platform may need more realistic promised times or additional partner capacity. If restaurant delay dominates for a small number of restaurants, targeted process coaching may be more effective than a network-wide intervention.

7.2 Caution on Interpretation

The chart reports recorded reasons, not necessarily the full causal chain. A customer may select long delivery time even when the underlying cause was traffic or restaurant delay. Therefore, cancellation reasons should be used as an operational signal and combined with the time and context fields.

8 Exploratory Analysis III: Orders by Hour

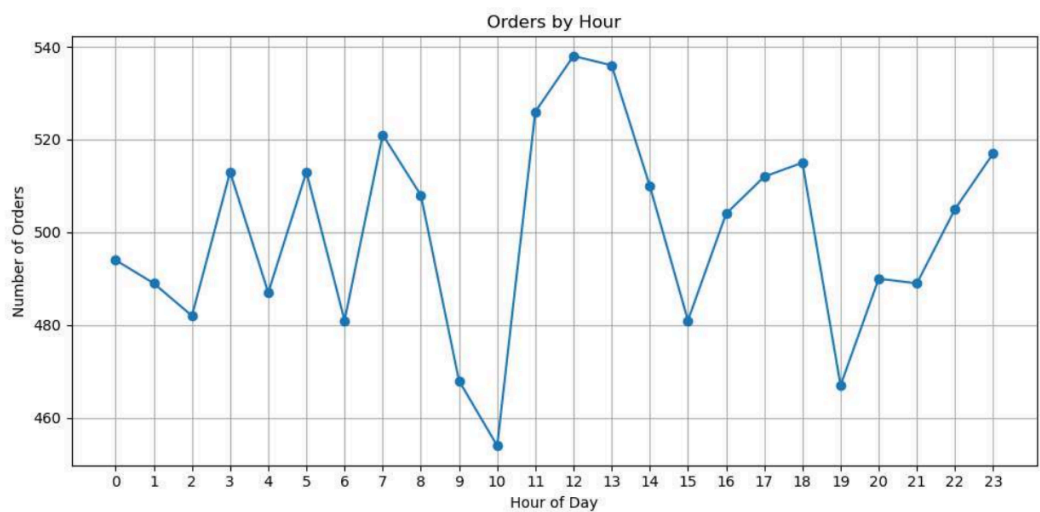


图 3 Hourly order volume from 00:00 to 23:00.

The hourly line chart shows order volume across a complete 24-hour cycle. Demand fluctuates rather than following a perfectly smooth pattern. The highest visible values occur around midday, particularly near hours 12 and 13, while the lowest visible point is around hour 10. The pattern indicates that capacity planning should be time-sensitive. Hourly demand is relevant to both restaurants and delivery partners. Restaurants may need additional kitchen staffing around peak order periods, while the platform may need to schedule sufficient delivery-partner availability before demand increases. The chart also supports the design of time-based service-level targets.

8.1 Practical Use in Operations

A useful next step would be to calculate hourly cancellation rate alongside hourly order volume. High volume with a low cancellation rate indicates healthy scale. High volume with a high cancellation rate indicates a capacity or execution problem. Low volume with a high cancellation rate may point to special conditions such as weather or poor partner coverage.

8.2 SQL and Python Alignment

The Python code creates `hourly_orders` using a group-by operation on `Order_Hour`. The SQL equivalent groups by `HOUR(Order_Time)` and orders by total orders in descending order. Keeping these definitions aligned is essential; otherwise, the Python chart and Power BI dashboard may tell different stories.

9 Exploratory Analysis IV: Traffic and Delivery Time

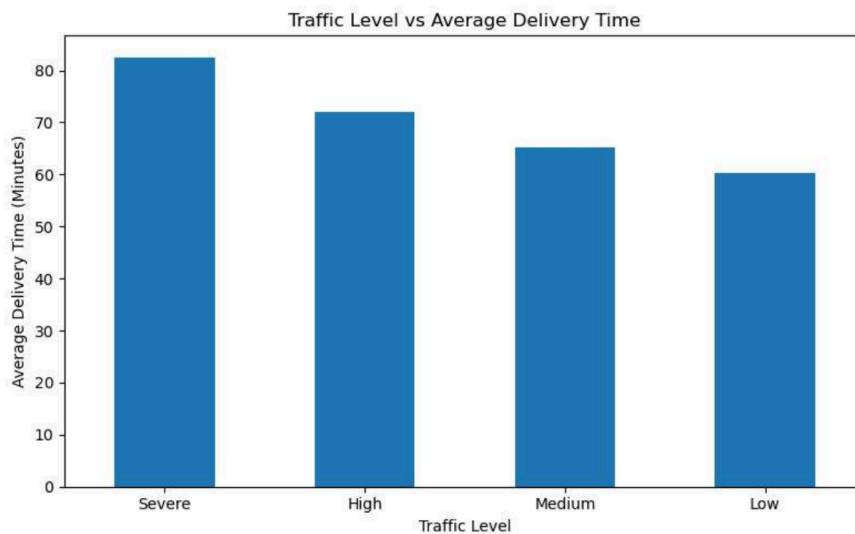


图 4 Average delivery time by traffic level.

Traffic is the clearest relationship shown in the supplied charts. Average delivery time rises from approximately 60 minutes under low traffic to approximately 65 minutes under medium traffic, approximately 72 minutes under high traffic,

and approximately 82 minutes under severe traffic. The difference between severe and low traffic is roughly twenty-two minutes, which is operationally significant.

The relationship is directionally intuitive: more congested roads increase travel time and reduce the predictability of arrival. However, preparation time may also contribute to the observed average. The supplied SQL therefore calculates both average delivery time and average food-preparation time by traffic level, allowing the team to separate road-related delay from kitchen-related delay.

9.1 Operational Actions

Traffic-sensitive promise times can reduce customer disappointment. The platform can also pre-position delivery partners, prioritize dispatch for orders that are approaching their expected handoff time, and use area-level traffic monitoring to adjust estimated delivery time. These actions should be evaluated against cancellation rate and customer experience.

9.2 Limitation

The chart is descriptive. It does not prove that traffic alone creates the full difference because traffic conditions may coincide with particular hours, areas, restaurants, or weather conditions. A segmented analysis or regression model would be required to estimate the independent contribution of traffic.

10 Exploratory Analysis V: Weather and Cancellation Rate

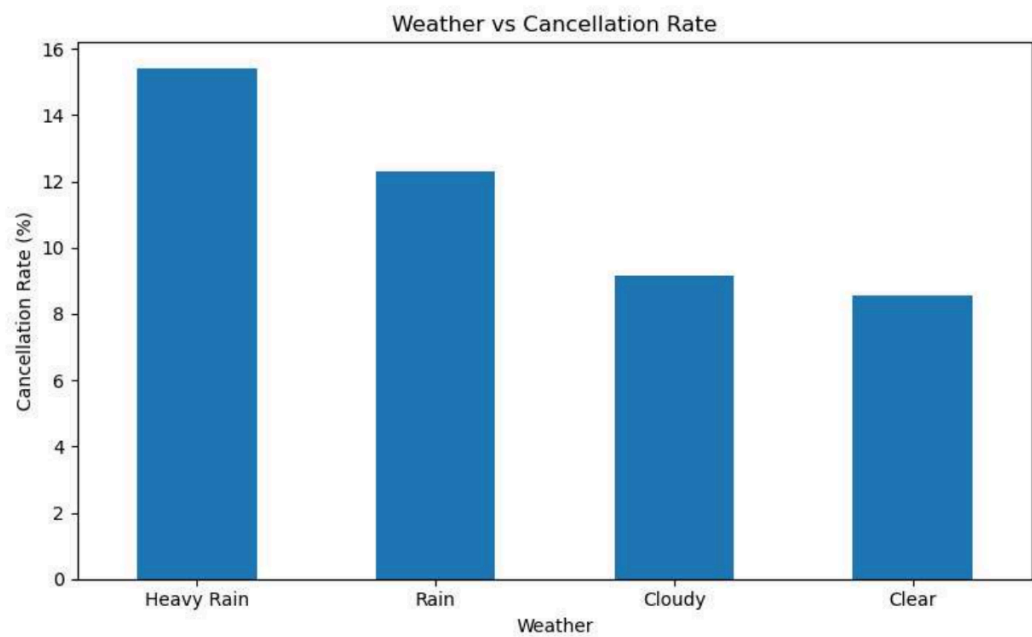


图 5 Cancellation rate by weather condition.

The weather chart shows the highest cancellation rate under heavy rain, at approximately 15%. Rain is next at roughly 12%, while cloudy and clear conditions are below 10% in the supplied visual. The pattern suggests that adverse weather increases delivery friction and customer uncertainty. Weather may affect road speed, partner willingness to accept orders, restaurant preparation, and customer decisions. Heavy rain can therefore create a compound operational effect. It is useful to monitor weather together with traffic because severe weather can also increase congestion.

10.1 Business Interpretation

The platform can prepare for adverse-weather periods by increasing partner incentives, communicating realistic delivery estimates, allowing customers to track order progress more clearly, and prioritizing restaurant-to-partner handoff. The goal is not only to reduce cancellations but also to protect delivery-partner safety and maintain service reliability.

10.2 Measurement Recommendation

The dashboard should display both the cancellation rate and the number of orders for each weather category. A high percentage based on very few orders should not receive the same operational priority as a similar percentage based on a large order volume.

11 SQL Database Implementation

The supplied SQL creates a dedicated database named `food_delivery_analytics` and a table named `food_delivery_orders`. This establishes a clean analytical workspace and documents the expected schema before queries are executed.

```
CREATE DATABASE food_delivery_analytics;
USE food_delivery_analytics;

CREATE TABLE food_delivery_orders (
  Order_ID VARCHAR(20),
  Order_Date DATE,
  Order_Time TIME,
  Customer_Area VARCHAR(100),
  Restaurant VARCHAR(100),
  Order_Value_INR DECIMAL(10,2),
  Food_Prep_Time_Min INT,
  Distance_KM DECIMAL(5,2),
  Delivery_Time_Min INT,
  Delivery_Partner VARCHAR(20),
  Order_Status VARCHAR(20),
  Cancellation_Reason VARCHAR(100),
  Weather VARCHAR(30),
  Traffic_Level VARCHAR(20)
);
```

The table definition is explicit and readable. It makes the data contract visible to the analyst and provides a foundation for validation. The use of decimal precision for order value and distance is appropriate because both fields may contain fractional values.

11.1 Validation Queries

The first validation queries inspect the schema and sample rows. `DESCRIBE food_delivery_orders` confirms the field structure, while `SELECT * ... LIMIT 10` provides a quick inspection of loaded data. `COUNT(*)` confirms the number of records available for analysis.

```
DESCRIBE food_delivery_orders;
SELECT COUNT(*) AS Total_Orders
FROM food_delivery_orders;
SELECT *
FROM food_delivery_orders
LIMIT 10;
```

12 SQL Metrics and KPI Definitions

The SQL stage calculates the key performance indicators used by the project. Each metric should be defined before it is placed on a dashboard so that the meaning remains stable across Python, SQL, and Power BI.

KPI	Definition	Population
Total orders	Count of all order records.	All orders.
Delivered orders	Count where status equals Delivered.	Delivered orders.
Cancelled orders	Count where status equals Cancelled.	Cancelled orders.
Cancellation rate	Cancelled orders divided by total orders multiplied by 100.	All orders.
Average delivery time	Mean delivery duration in minutes.	Delivered orders only.
Average order value	Mean order value in INR.	All orders in supplied SQL.
Delivered revenue	Sum of order value for delivered orders.	Delivered orders only.

12.1 Status-Based Measures

The delivered and cancelled counts are calculated using explicit filters. The cancellation rate uses a conditional sum divided by the total count, rounded to two decimal places. This is preferable to dividing by delivered orders because the business question is the share of all orders that were cancelled.

```
SELECT COUNT(*) AS Delivered_Orders
FROM food_delivery_orders
WHERE Order_Status = 'Delivered';

SELECT COUNT(*) AS Cancelled_Orders
FROM food_delivery_orders
WHERE Order_Status = 'Cancelled';

SELECT ROUND(
  SUM(CASE WHEN Order_Status = 'Cancelled' THEN 1 ELSE 0 END) * 100.0
  / COUNT(*), 2
) AS Cancellation_Rate_Percent
FROM food_delivery_orders;
```

12.2 Delivery and Revenue Measures

The average delivery time is filtered to delivered orders because cancelled orders do not represent a completed delivery. Delivered revenue is also filtered to delivered orders, which avoids treating cancelled order values as realized revenue.

13 SQL Segmentation Queries

The project uses SQL to segment performance by cancellation reason, area, hour, traffic, weather, and restaurant. These segments convert a single KPI into actionable diagnostic views.

13.1 Cancellation Reason Ranking

```
SELECT Cancellation_Reason,
       COUNT(*) AS Cancelled_Orders
FROM food_delivery_orders
WHERE Order_Status = 'Cancelled'
```

```
GROUP BY Cancellation_Reason
ORDER BY Cancelled_Orders DESC;
```

This query produces the ranking visualized in the cancellation chart. Sorting in descending order makes the largest improvement opportunity appear first.

13.2 Hourly Demand and Cancellation

```
SELECT HOUR(Order_Time) AS Order_Hour,
       COUNT(*) AS Total_Orders,
       SUM(CASE WHEN Order_Status = 'Cancelled' THEN 1 ELSE 0 END)
       AS Cancelled_Orders,
       ROUND(
         SUM(CASE WHEN Order_Status = 'Cancelled' THEN 1 ELSE 0 END)
         * 100.0 / COUNT(*), 2
       ) AS Cancellation_Rate
FROM food_delivery_orders
GROUP BY HOUR(Order_Time)
ORDER BY Order_Hour;
```

This query is particularly valuable because it combines demand and reliability. An hour with high order volume is not automatically a good hour if cancellations also rise.

13.3 Traffic and Weather Segmentation

Traffic analysis compares average delivery time and average preparation time. Weather analysis calculates total orders, cancelled orders, and cancellation rate for each weather category. The use of CASE inside the aggregation keeps the rate calculation transparent.

14 Restaurant and Area Performance

Restaurant-level analysis is useful because restaurant delay is one of the leading cancellation categories in the supplied chart. The project query calculates total orders, average preparation time, average delivery time, and cancellation rate for each restaurant.

```
SELECT Restaurant,
       COUNT(*) AS Total_Orders,
       ROUND(AVG(Food_Prep_Time_Min), 2) AS Avg_Preparation_Time,
       ROUND(AVG(Delivery_Time_Min), 2) AS Avg_Delivery_Time,
       ROUND(
         SUM(CASE WHEN Order_Status = 'Cancelled' THEN 1 ELSE 0 END)
         * 100.0 / COUNT(*), 2
       ) AS Cancellation_Rate
FROM food_delivery_orders
GROUP BY Restaurant
ORDER BY Cancellation_Rate DESC;
```

This query helps distinguish restaurants with slow preparation from restaurants whose delays occur after handoff. A restaurant with high preparation time and high cancellation rate may require kitchen-process support. A restaurant with normal preparation time but high delivery time may instead reflect distance, traffic, or partner allocation.

14.1 Area-Level Comparison

Area-level average delivery time should be read together with order count and average distance. A small area can appear slow because of a few outliers. The dashboard can reduce this risk by displaying the number of delivered orders as a tooltip or secondary measure.

14.2 Suggested Priority Matrix

Observed condition	Likely focus	Suggested action
High preparation, high cancellation	Restaurant operations	Review acceptance, kitchen throughput, and handoff readiness.
Normal preparation, high delivery time	Dispatch or road conditions	Review distance, traffic, and partner allocation.
High volume, rising hourly cancellation	Peak capacity	Increase partner coverage and adjust promised times.
High weather cancellation	Adverse-weather resilience	Use weather-aware capacity and customer communication.

15 Power BI Dashboard Architecture

The supplied PBIX workbook contains one report page and eleven visual containers. The theme is named Red_Black_Premium_Theme, giving the dashboard a dark, high-contrast visual identity. The dashboard combines summary cards with comparative charts rather than presenting a long table of raw records.

Visual family	Count	Purpose in the dashboard
Card visuals	3	Present headline values or identifiers.
Clustered bar charts	4	Compare categories such as area, restaurant, reason, and traffic.
Clustered column charts	2	Compare order value and delivery time by category.
Column chart	1	Compare weather categories with order status series.
Line chart	1	Show hourly order volume.

15.1 Dashboard Visual Inventory

The layout includes a line chart for count of order ID by order hour; bar charts for cancellation reason, area delivery time, restaurant delivery time, and traffic-level order count; column charts for weather/status and average delivery time by traffic; and a chart for average order value by customer area. Three card visuals are also present.

15.2 Dashboard Strengths

The dashboard covers the main dimensions identified in the SQL documentation. It gives the viewer a path from demand to performance: when orders arrive, why they are cancelled, where deliveries are slow, how traffic affects time, and how weather relates to order status. The use of multiple chart types also creates visual variety while keeping each visual aligned with its analytical purpose.

15.3 Dashboard Improvement Opportunities

The dashboard would be stronger if the three card visuals displayed explicit titles and meaningful KPI labels. Suggested cards are Total Orders, Cancellation Rate, and Average Delivered Delivery Time. A date slicer, status slicer, traffic slicer, and weather slicer would improve exploration. Tooltips should include record count and the denominator used for each rate.

16 Integrated Findings

The strongest integrated finding is that service reliability is influenced by several connected operational conditions. Severe traffic is associated with much longer delivery times, heavy rain is associated with a higher cancellation rate, and long delivery time is the leading cancellation reason. These observations form a coherent operational chain: adverse road conditions can increase delivery time, which may then increase cancellation risk.

The second finding is that area differences are comparatively narrow in the supplied chart. This suggests that the platform's broad geographic service level is relatively consistent, or that area-level averages are being shaped by common system-wide factors. To find meaningful improvement opportunities, management should combine area with restaurant, distance, traffic, and hour.

The third finding is that demand is distributed throughout the day, with a visible midday peak. Staffing and partner planning should therefore be based on time windows rather than daily totals. A daily average can hide the capacity pressure created by a concentrated peak.

The fourth finding is that cancellation reasons can be grouped into operational families. Long delivery time, partner unavailability, and restaurant delay are execution-related. Customer changed mind and payment issue are customer or transaction-related. This grouping can help assign accountability and design targeted interventions.

17 Recommendations and Action Plan

The recommendations below are designed to connect the analytical findings with operational decisions. They should be piloted, measured, and refined rather than treated as guaranteed outcomes.

17.1 Immediate Actions

The first immediate action is to standardize KPI definitions across Python, SQL, and Power BI. The dashboard should clearly distinguish total orders, delivered orders, cancelled orders, cancellation rate, and delivered revenue. The second is to add explicit titles to all card visuals and to display the unit for every metric. The third is to add order count beside every rate so that percentages are interpreted with their denominator.

17.2 Near-Term Actions

The platform should create traffic-aware estimated delivery times and monitor severe-traffic periods separately. Partner availability should be compared with hourly demand, especially around the midday peak. Restaurants with high preparation time or high cancellation rate should receive a focused review of acceptance and handoff processes.

17.3 Weather Readiness

Weather-aware planning should begin before heavy rain periods. The platform can increase partner coverage where safe, communicate realistic delivery times, and surface live status updates to customers. Monitoring should include cancellation rate, average delivery time, partner acceptance, and order volume by weather category.

17.4 Measurement Framework

Initiative	Primary KPI	Supporting measures
Traffic-aware ETA	Average delivered delivery time	Long-delivery cancellations, customer complaints, on-time share.
Peak staffing	Hourly cancellation rate	Order volume, partner availability, preparation time.
Restaurant support	Restaurant cancellation rate	Preparation time, handoff delay, delivery time.
Weather playbook	Weather cancellation rate	Traffic, partner acceptance, order volume.

18 Limitations and Future Scope

The current project is a descriptive analytics implementation. Several limitations should be stated clearly in an academic submission. First, the report relies on the supplied charts and SQL documentation rather than a separately attached raw CSV; therefore, exact row-level recalculation is outside the scope of this document. Second, the charts show averages and rates, but not confidence intervals or distributions. Third, relationships between traffic, weather, distance, restaurant, and time may be confounded.

The Power BI layout also contains visuals whose titles or measures appear to require refinement. In particular, the layout summary indicates that some visuals use Sum projections for fields described as averages. This should be validated in the PBIX model and corrected with explicit average measures where appropriate. A professional dashboard should ensure that the visual title, aggregation function, and displayed unit all agree.

18.1 Future Enhancements

A future version could add a star schema with a fact-orders table and dimensions for date, restaurant, area, weather, and traffic. It could include a calendar table, a proper measures layer, drill-through pages, bookmarks, and conditional formatting. Predictive extensions could forecast hourly demand, estimate late-delivery risk, or classify likely cancellations. Route-level data could support optimization, while customer-level history could support retention analysis.

18.2 Governance and Reproducibility

The project should retain the cleaned dataset, EDA notebook, SQL script, PBIX workbook, chart exports, and this report in a versioned folder. Each refresh should record the data date, row count, data-quality checks, and any changes to KPI definitions. This practice makes the project easier to explain, evaluate, and maintain.

19 Conclusion

This food-delivery analytics project demonstrates how a raw operational dataset can be converted into a structured decision-support product. The EDA stage reveals distributional patterns and produces explanatory charts. The SQL stage formalizes the business logic and creates repeatable metrics. The Power BI stage packages the results into a dashboard that can be reviewed by a mentor or business stakeholder.

The most important operational signals are the connection between traffic and delivery time, the elevated cancellation rate during heavy rain, the dominance of long delivery time among cancellation reasons, and the midday variation in order volume. Area differences are present but relatively narrow in the supplied chart, so area-level action should be supported by distance, restaurant, traffic, and time analysis.

Overall, the project provides a strong foundation for a larger analytics system. With clearer KPI cards, validated average measures, richer filters, and a documented refresh process, the current dashboard can evolve from an academic submission into a practical operational monitoring tool.

20 Appendix A: Core SQL Script

```
CREATE DATABASE food_delivery_analytics;
USE food_delivery_analytics;

CREATE TABLE food_delivery_orders (
    Order_ID VARCHAR(20),
    Order_Date DATE,
    Order_Time TIME,
    Customer_Area VARCHAR(100),
    Restaurant VARCHAR(100),
    Order_Value_INR DECIMAL(10,2),
    Food_Prep_Time_Min INT,
    Distance_KM DECIMAL(5,2),
```

```
Delivery_Time_Min INT,  
Delivery_Partner VARCHAR(20),  
Order_Status VARCHAR(20),  
Cancellation_Reason VARCHAR(100),  
Weather VARCHAR(30),  
Traffic_Level VARCHAR(20)  
);  
  
SELECT COUNT(*) AS Total_Orders  
FROM food_delivery_orders;  
  
SELECT ROUND(AVG(Delivery_Time_Min), 2) AS Avg_Delivery_Time_Min  
FROM food_delivery_orders  
WHERE Order_Status = 'Delivered';  
  
SELECT ROUND(AVG(Order_Value_INR), 2) AS Avg_Order_Value  
FROM food_delivery_orders;  
  
SELECT ROUND(SUM(Order_Value_INR), 2) AS Total_Revenue  
FROM food_delivery_orders  
WHERE Order_Status = 'Delivered';
```

21 Appendix B: EDA Plotting Logic

```
# Orders by Hour  
hourly_orders = df.groupby("Order_Hour").size()  
hourly_orders.plot(kind="line", marker="o")  
  
# Cancellation Reasons  
cancelled = df[df["Order_Status"] == "Cancelled"]  
cancelled["Cancellation_Reason"].value_counts().plot(kind="bar")  
  
# Traffic vs Delivery Time  
traffic_delivery = (  
    df[df["Order_Status"] == "Delivered"]  
    .groupby("Traffic_Level")["Delivery_Time_Min"]  
    .mean()  
    .sort_values(ascending=False)  
)  
traffic_delivery.plot(kind="bar")  
  
# Area-wise Delivery Time  
area_delivery = (  
    df[df["Order_Status"] == "Delivered"]  
    .groupby("Customer_Area")["Delivery_Time_Min"]  
    .mean()  
    .sort_values(ascending=False)  
)  
area_delivery.plot(kind="bar")  
  
# Weather vs Cancellation Rate
```

```

weather_cancel = (
    df.groupby("Weather")
      .apply(lambda x: (x["Order_Status"] == "Cancelled").mean() * 100)
      .sort_values(ascending=False)
)
weather_cancel.plot(kind="bar")

```

22 Appendix C: Source Inventory and Submission Checklist

Source file	Role in the project	Included
documaentofproject.txt	SQL schema, KPI queries, segmentation queries, and EDA plotting code.	Yes
area_delivery_time(1).pdf	Five supplied charts used as figures and interpreted in this report.	Yes
newporject1.pbix	Power BI dashboard layout, visuals, and theme evidence.	Yes
Generated PDF report	Final academic narrative connecting the workflow and findings.	Yes

22.1 Presentation Checklist

Before submission, the student should fill in the name, institute, course, mentor, and signature fields on the title and declaration pages. The PBIX should be opened once to verify that the three card visuals have meaningful titles and that all measures use the intended aggregation. The final report should be submitted together with the PBIX and, where permitted, the cleaned dataset or EDA notebook.

22.2 References to Supplied Project Materials

- [1] User-provided SQL and EDA documentation, documaentofproject.txt.
- [2] User-provided chart collection, area_delivery_time(1).pdf.
- [3] User-provided Power BI workbook, newporject1.pbix.

End of Report

23 Power BI Dashboard Screenshot

The following page presents the final Power BI dashboard supplied for this project. It brings together the headline order KPIs, traffic-level order counts, cancellation reasons, area-wise delivery time, hourly demand, restaurant delivery time, area-wise order value, weather and order status, and traffic-level delivery time in a single management view.

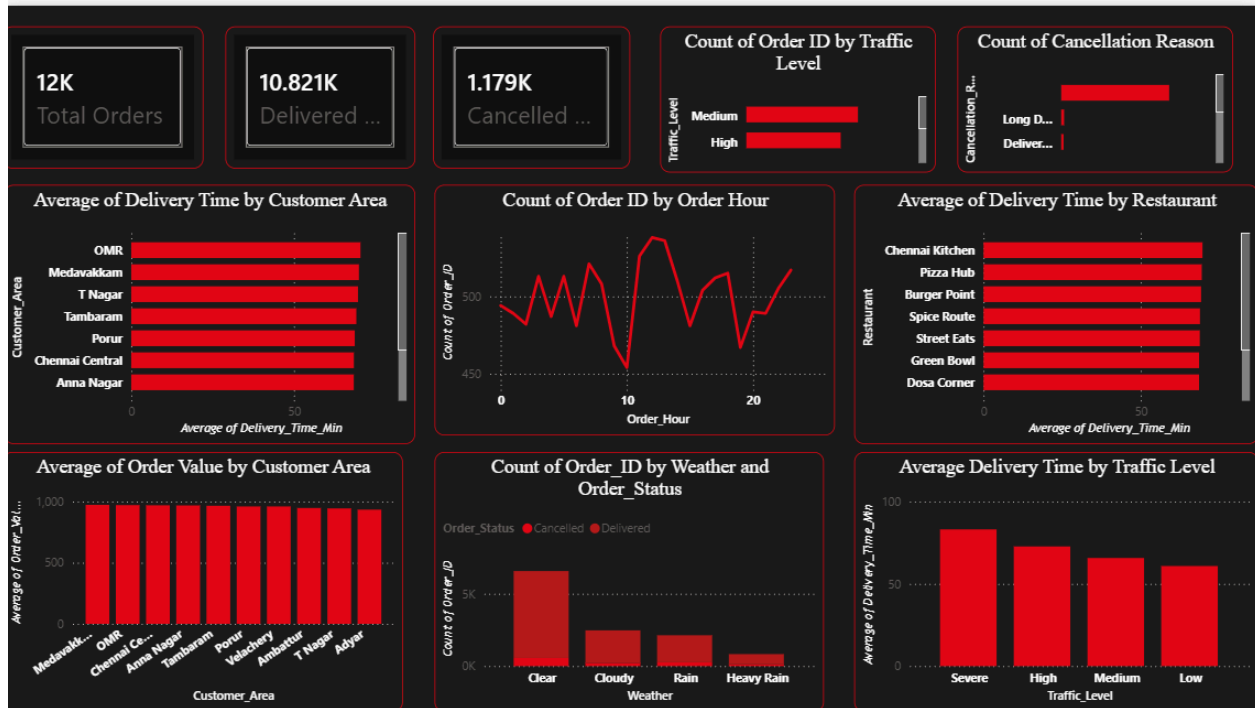


图 6 Final Power BI dashboard view.